SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

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COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO* (BL3, BL6)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **TOTAL MARKS: 25**

Annexure A
Comparative Analysis

Code of Conduct:

I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *E. Suchier*

Annexure A
Comparative Analysis

5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

In IPSec**,** key management is essential for establishing secure communication between two endpoints. The two main approaches are manual key management and automated key management using protocols such as IKE (Internet Key Exchange)**.**

Aspect	Manual Key Management	Automated Key Management (IKE)
Key generation	Keys are manually created and configured by administrators.	Keys are automatically negotiated and generated using IKE.
Security	More prone to human errors, weak keys, and accidental exposure.	Provides stronger security through automatic key negotiation, authentication, and periodic rekeying.
Scalability	Poor scalability because every device must be configured individually.	Highly scalable; suitable for large networks with many VPN endpoints.
Key renewal	Keys must be manually replaced or updated.	Keys can be automatically renewed before expiration.
Operational complexity	High administrative effort and configuration overhead.	Lower operational effort after initial deployment and configuration.
Risk of human error	High, especially in large organizations.	Lower because key generation and exchange are automated.
Deployment	Practical mainly for small networks or temporary configurations.	Suitable for enterprise VPNs, site-to-site VPNs, and remote-access environments.
Failure management	Troubleshooting and replacing keys is manual.	IKE can automatically renegotiate keys and establish new security associations.

Impact on Large Organizations

Security: Manual key management introduces significant security risks because administrators must generate, distribute, store, and periodically replace keys. A compromised or incorrectly configured key can expose communications. IKE improves security by automating authentication and cryptographic key negotiation and supporting regular rekeying.

Scalability: Manual management becomes impractical as the number of IPSec endpoints increases. For example, managing keys individually across hundreds or thousands of VPN gateways creates a substantial administrative burden. IKE scales much better because devices can automatically negotiate keys and establish Security Associations (SAs).

Operational Complexity: Manual management requires continuous administrator involvement for key configuration, rotation, replacement, and troubleshooting. Automated management significantly reduces these tasks. Although IKE requires an initial configuration of authentication mechanisms, policies, and parameters, the ongoing management is considerably simpler.

Conclusion

For a large organization, automated key management using IKE is strongly recommended**.** Manual key management may be acceptable for small, static, or laboratory environments, but it does not scale efficiently and has a higher risk of human error. IKE provides better security, scalability, automatic key renewal, and lower operational overhead, making it the preferred approach for enterprise IPSec deployments.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand